WORKED SOLUTION

STEP III 2004, Question 8

Pure • Differential Equations

STEP III 2004, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q8

The key idea

The equation is not linear in y, because of the $1/y$ term. But multiplying through by y turns $y\,\frac{\mathrm{d}y}{\mathrm{d}x}$ into half the derivative of y^2, and the $1/y$ term becomes a plain function of x. So in terms of $u = y^2$ the equation is linear (an equation of Bernoulli type), and an integrating factor finishes it off. The three curves are then conics, and identifying them makes the sketch easy.

Solution

The substitution

Let $u = y^2$. By the chain rule,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 2y\left(f(x)\,y + \frac{g(x)}{y}\right) = 2f(x)\,y^2 + 2g(x).$$

Replacing y^2 by u,

$$\frac{\mathrm{d}u}{\mathrm{d}x} - 2f(x)\,u = 2g(x),$$

which is linear in u: u and $\frac{\mathrm{d}u}{\mathrm{d}x}$ appear only to the first power, with coefficients depending on x alone. This is what we had to show.

Solving the equation

Here $f(x) = \frac{1}{x}$ and $g(x) = -1$, so the linear equation is

$$\frac{\mathrm{d}u}{\mathrm{d}x} - \frac{2}{x}\,u = -2.$$

The integrating factor is $\exp\left(-\int \frac{2}{x}\,\mathrm{d}x\right) = \mathrm{e}^{-2\ln|x|} = \dfrac{1}{x^2}$. Multiplying through by it,

$$\frac{1}{x^2}\frac{\mathrm{d}u}{\mathrm{d}x} - \frac{2}{x^3}\,u = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{u}{x^2}\right) = -\frac{2}{x^2}.$$

Integrating, $\dfrac{u}{x^2} = \dfrac{2}{x} + C$ for a constant C, so the general solution is

$$y^2 = 2x + Cx^2.$$

(The original equation needs $x \neq 0$ and $y \neq 0$, so each solution lives on an interval of x not containing 0 on which $2x + Cx^2 > 0$, with y of one sign throughout.)

The three curves

Each of the three points has $x > 0$ and $y > 0$, so each solution curve is the part of $y^2 = 2x + Cx^2$ with $y > 0$ on the interval of positive x containing the point.

Through $(1,1)$**:** $1 = 2 + C$, so $C = -1$ and $y^2 = 2x - x^2$. Completing the square,

$$(x-1)^2 + y^2 = 1,$$

the circle with centre $(1,0)$ and radius 1. The solution curve is the upper semicircle $y = \sqrt{2x - x^2}$ for $0 < x < 2$.

Through $(2,2)$**:** $4 = 4 + 4C$, so $C = 0$ and $y^2 = 2x$: the parabola with vertex at the origin, axis along the x-axis. The solution curve is $y = \sqrt{2x}$ for $x > 0$.

Through $(4,4)$**:** $16 = 8 + 16C$, so $C = \frac{1}{2}$ and $y^2 = 2x + \frac{1}{2}x^2$. Completing the square, $\frac{1}{2}(x+2)^2 - y^2 = 2$, that is

$$\frac{(x+2)^2}{4} - \frac{y^2}{2} = 1,$$

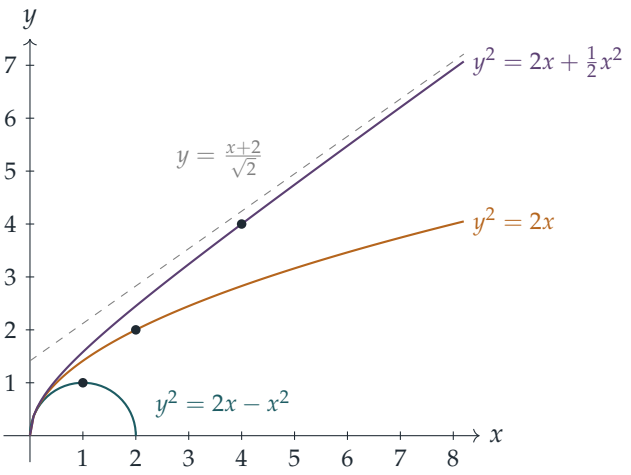
a hyperbola with centre $(-2,0)$, vertices $(0,0)$ and $(-4,0)$, and asymptotes $y = \pm\frac{1}{\sqrt{2}}(x+2)$. The solution curve is the upper half of the right-hand branch, $y = \sqrt{2x + \frac{1}{2}x^2}$ for $x > 0$.

> **Ask yourself** Why not include the left-hand branch of the hyperbola, or the lower halves?

A solution of the differential equation must be differentiable on an interval and satisfy the equation there, and the equation is undefined when $x = 0$ or $y = 0$. Starting at $(4,4)$ and moving left, the curve reaches $(0,0)$, where both fail, so it cannot be continued through. The left branch (with $x \leq -4$) and the lower halves satisfy $y^2 = 2x + \frac{1}{2}x^2$ but are different solutions, not the one through $(4,4)$.

Sketch

All three curves start at the origin with a vertical tangent (near $x = 0$ each behaves like $y \approx \sqrt{2x}$), and at the given points their gradients are $\frac{1}{1} - \frac{1}{1} = 0$, $\frac{2}{2} - \frac{1}{2} = \frac{1}{2}$ and $\frac{4}{4} - \frac{1}{4} = \frac{3}{4}$. For larger x the circle comes back down to $(2,0)$, the parabola grows like $\sqrt{x}$, and the hyperbola approaches its asymptote $y = \frac{1}{\sqrt{2}}(x+2)$. Since $2x - x^2 < 2x < 2x + \frac{1}{2}x^2$ for $x > 0$, the circle lies below the parabola, which lies below the hyperbola.



Checks

- Substitute back: if $y^2 = 2x + Cx^2$ then $2yy' = 2 + 2Cx$, so $y' = \frac{1+Cx}{y}$, while $\frac{y}{x} - \frac{1}{y} = \frac{y^2-x}{xy} = \frac{x+Cx^2}{xy} = \frac{1+Cx}{y}$. They agree.
- Stepping the original equation numerically (Runge–Kutta) from $(4, 4)$ to $x = 8$ gives $y \approx 6.92820$, matching $\sqrt{16+32} = \sqrt{48}$; similar runs from $(1, 1)$ and $(2, 2)$ match the semicircle and parabola to about 10^{-12}.

Common slips

- Getting the integrating factor as x^2 instead of x^{-2}: the coefficient of u is $-\frac{2}{x}$, so the exponent is $-2\ln x$.
- Forgetting the factor 2 from $\frac{du}{dx} = 2y\frac{dy}{dx}$, which leads to $y^2 = x + Cx^2$; that fails the back-substitution check.
- Sketching the whole circle, both halves of the parabola or both branches of the hyperbola as "the solution curve".

Beyond STEP

The trick here works for any *Bernoulli equation* $y' = f(x)y + g(x)y^n$: the substitution $u = y^{1-n}$ always makes it linear (here $n = -1$, so $u = y^2$). The care needed at $x = 0$ and $y = 0$ is a first glimpse of the existence and uniqueness theorem: through each point where the right-hand side is well behaved there is exactly one solution, and the family of curves can only meet at points where it isn't, as all of ours do at the origin. Boyce and DiPrima, *Elementary Differential Equations*, covers Bernoulli equations and this theorem.

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